EXP 30

```
clc;
clear;
close all;

% Cellular Frequency Planning and Handover Performance
% under User Mobility Conditions

% Step 1: Define system parameters
numCells = 7;
cellRadius = 500;            % Cell radius in meters
reuseFactor = 3;             % Frequency reuse factor
numChannels = 60;            % Total available channels
handoverThreshold = -85;     % Handover threshold in dBm
hysteresis = 5;              % Handover hysteresis in dB

% Step 2: Create cell locations
angles = (0:5)*pi/3;
cellX = [0, 2*cellRadius*cos(angles)];
cellY = [0, 2*cellRadius*sin(angles)];

% Frequency group assignment
frequencyGroups = mod(0:numCells-1,reuseFactor) + 1;

% Step 3: Simulate mobile user movement
numSteps = 300;
userX = linspace(-900,900,numSteps);
userY = 100*sin(linspace(0,2*pi,numSteps));

% Path-loss model parameters
```

```
Pt = 30;                % Transmit power in dBm
PL0 = 40;               % Reference path loss in dB
d0 = 1;                 % Reference distance in meters
pathLossExponent = 3.5;

servingCell = zeros(1,numSteps);
signalStrength = zeros(numCells,numSteps);
handoverCount = 0;
successfulHandovers = 0;
failedHandovers = 0;

currentCell = 1;

% Step 4: Calculate received signal strength
for t = 1:numSteps

   for c = 1:numCells
      distance = sqrt((userX(t)-cellX(c))^2 + ...
                  (userY(t)-cellY(c))^2);
      distance = max(distance,d0);

      pathLoss = PL0 + 10*pathLossExponent*log10(distance/d0);

      % Log-normal shadow fading
      shadowFading = 4*randn;

      signalStrength(c,t) = Pt - pathLoss + shadowFading;
   end

   % Find strongest cell
   [bestRSSI,bestCell] = max(signalStrength(:,t));
```

```
    % Handover decision with hysteresis
    if currentCell ~= bestCell
        currentRSSI = signalStrength(currentCell,t);

        if bestRSSI > currentRSSI + hysteresis && ...
                currentRSSI < handoverThreshold
            handoverCount = handoverCount + 1;

            % Handover admission check (simplified)
            if rand > 0.05
                currentCell = bestCell;
                successfulHandovers = successfulHandovers + 1;
            else
                failedHandovers = failedHandovers + 1;
            end
        end
    end

    servingCell(t) = currentCell;
end

% Step 5: Calculate performance metrics
handoverSuccessRate = 100 * successfulHandovers / ...
                    max(handoverCount,1);

handoverFailureRate = 100 * failedHandovers / ...
                    max(handoverCount,1);

% Step 6: Display results
fprintf('\nCELLULAR FREQUENCY PLANNING\n');
```

```
fprintf('Number of cells = %d\n',numCells);
fprintf('Cell radius = %d meters\n',cellRadius);
fprintf('Frequency reuse factor = %d\n',reuseFactor);
fprintf('Total available channels = %d\n',numChannels);
fprintf('Channels per reuse group (approx.) = %d\n', ...
    floor(numChannels/reuseFactor));

fprintf('\nHANDOVER PERFORMANCE\n');
fprintf('Total handover attempts = %d\n',handoverCount);
fprintf('Successful handovers = %d\n',successfulHandovers);
fprintf('Failed handovers = %d\n',failedHandovers);
fprintf('Handover success rate = %.2f%%\n',handoverSuccessRate);
fprintf('Handover failure rate = %.2f%%\n',handoverFailureRate);

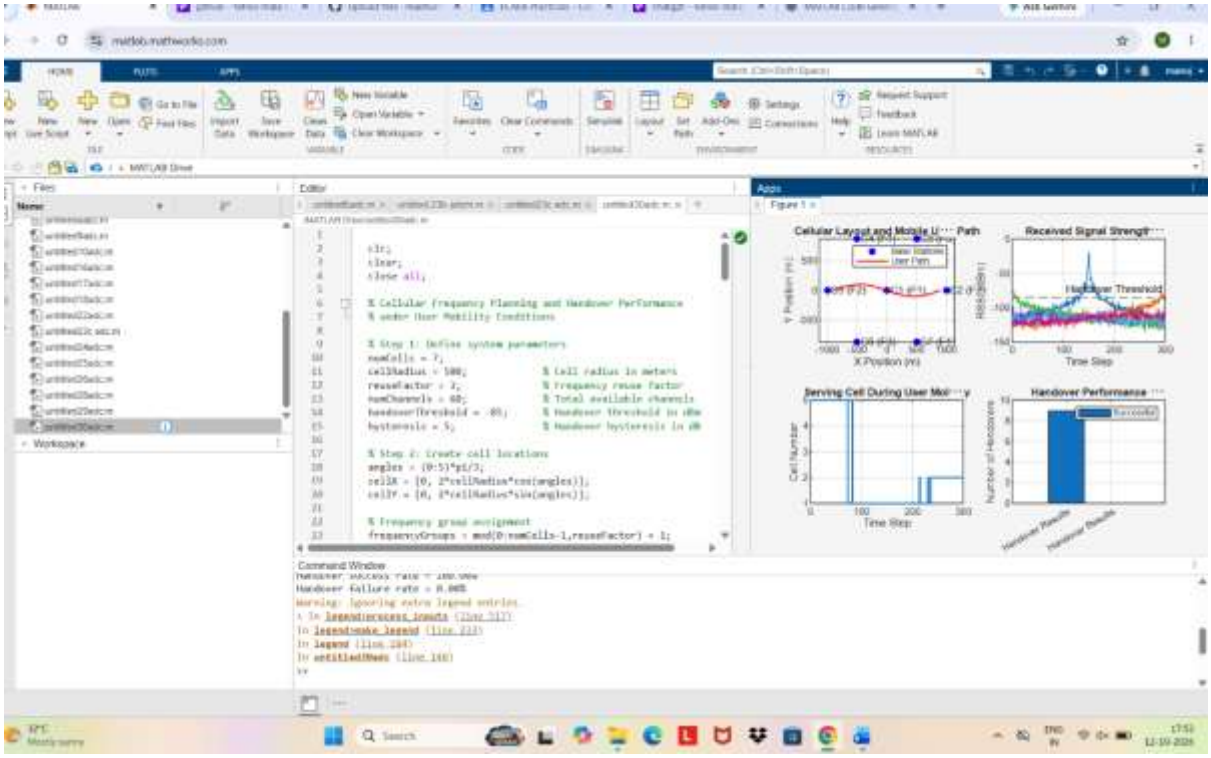
% Step 7: Plot cellular layout and mobile path
figure;

subplot(2,2,1);
plot(cellX,cellY,'bo','MarkerFaceColor','b');
hold on;
plot(userX,userY,'r-','LineWidth',1.5);
for c = 1:numCells
    text(cellX(c),cellY(c),sprintf(' C%d (F%d)', ...
        c,frequencyGroups(c)));
end
title('Cellular Layout and Mobile User Path');
xlabel('X Position (m)');
ylabel('Y Position (m)');
legend('Base Stations','User Path');
grid on;
axis equal;
```

```
% Step 8: Plot received signal strength
subplot(2,2,2);
plot(1:numSteps,signalStrength','LineWidth',1);
hold on;
yline(handoverThreshold,'k--','Handover Threshold');
title('Received Signal Strength');
xlabel('Time Step');
ylabel('RSSI (dBm)');
grid on;

% Step 9: Plot serving cell changes
subplot(2,2,3);
stairs(1:numSteps,servingCell,'LineWidth',1.5);
title('Serving Cell During User Mobility');
xlabel('Time Step');
ylabel('Cell Number');
yticks(1:numCells);
grid on;

% Step 10: Plot handover performance
subplot(2,2,4);
bar([successfulHandovers failedHandovers]);
set(gca,'XTickLabel',{'Handover Results'});
legend('Successful','Failed');
title('Handover Performance');
ylabel('Number of Handovers');
grid on;
```


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Cellular Layout and Mobile U··· Path
Received Signal Strengt···
Handover Threshold
Serving Cell During User Mo··· y
Handover Performance ···
Handovers
X Position (m)
Y Position (m)
Time Step
Cell Number
Number of Handovers
Command Window
Handover success rate = 100.00%
Handover failure rate = 0.00%
Warning: Ignoring extra legend entries.